MILITARY SPECIFICATION SHEET

ELECTRON TUBE, KLYSTRON

TYPE 2K45

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: CW, reflex oscillator, integral cavity, thermally tuned, frequency range 8,500 to 9,660 MHz

ABSOLUTE RATINGS: See note 1

Parameter:	Ef	Ers	Er	Pp	Ik1	Ehk	Ecc	Alt
Unit:	V	Vdc	Vdc	W	mAdc	Vdc	Vdc	ft
Maximum:	6.8	350	-350	7.0	30	±45	-75	10,000
Minimum:	5.8	---	---	---	---	---	---	---

PHYSICAL CHARACTERISTICS:

Dimensions:	See figure 1
Cathode:	Unipotential
Mounting:	See note 2

TEST CONDITIONS: (See notes 1 and 3)	Ef V	Ecc Vdc	Ers Vdc	Er Vdc	Ik1 mAdc	Ik2 mAdc	Rk2 Ohms
	6.3	-45 to +15	300	-60 to -175	22	0 to 25	750 ±5%

Frequency		
F	MHz	±
F1	8,500	0.3%
F2	9,080	0.3%
F3	9,660	0.3%

GENERAL:

Qualification - Required

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS MINIMUM	LIMITS MAXIMUM	UNIT
	Quality conformance inspection, part 1	4					
1256	Cathode current						
	Cathode (1)	1	Er = -150 Vdc; Ecc = -45 Vdc	Ik1	---	30	mAdc
	Cathode (2)	1	Ecc = +15 Vdc	Ik2	---	25	mAdc
1266	Total grid current	-	Ecc = -10 Vdc	Ic	0	-2.0	µAdc
4213	Reflector voltage	-	Er (Mode A) max Po	Er	-95	-145	Vdc
4214	Cathode emission						
	Cathode (1)	1	Er = -150 Vdc; Ecc = -45 Vdc; Ef = 5.8 V	ΔIk1 Ik1	---	15	%
	Cathode (2)	1	Ecc = +15 Vdc; Ef = 5.8 V	ΔIk2 Ik2	---	7.5	%
4229	Reflector-leakage current	5	Er = 150 Vdc; Ecc = +10 Vdc	Ir	---	3.0	µAdc
4250	Power output	2	Ef = 5.8 V; F = F1 to F3	Po	20	---	mW
---	Thermal tuning range	-	Er (Mode A) max Po; Ecc = -15 to -35 Vdc; Ef = 5.8 V				
			High frequency	F	9.660	---	MHz
			Low frequency	F	---	8.500	MHz
---	Thermal tuning time						
	Tuning time (1)	6	F = F1 to F3	t	---	9.0	sec
	Tuning time (2)	6	F = F3 to F1	t	---	9.0	sec
	Quality conformance inspection, part 2						
1031	High-frequency vibration	-	No voltages	---	---	---	---
1211	Insulation of electrodes	1	Ers = 300 Vdc; tube cold	Rk1-rs Rk2-rs Rh1-rs	2.0 2.0 2.0	--- --- ---	Meg Meg Meg
1256	Electrode current (cutoff)	-	Ecc = -45 Vdc	Ik2	---	0.5	mAdc
4280	Electronic tuning range	-	Er (Mode A)/50% max Po; F = F1 to F3	ΔF	35	---	MHz
4289	Heater current	-		If	710	815	mA
4270	Heater-cathode leakage	-	Ehk1 = ±45 Vdc Ehk2 = ±45 Vdc	Ihk1 Ihk2	--- ---	100 100	µAdc µAdc

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 3						
---	Life test	-	Group B: Ecc = 0; Rk2 = 800 ohms	t	500	---	hrs
---	Life-test end points:						
4250	Power output	2	Ef = 5.8 V; F = F1 to F3	Po	10	---	mW
---	Thermal tuning range	-	Er (Mode A)/max Po; Ecc = +15 to -35 Vdc; Ef = 5.8 V				
			High frequency	F	9,660	---	MHz
			Low frequency	F	---	8,500	MHz

NOTES:

1. Cathode (1) refers to parts associated with the rf section of the tube. Cathode (2) refers to parts associated with the thermal tuner section of the tube. Symbol Pp refers to power dissipation in the thermal tuner anode.

2. The tube shall be fixed firmly in a suitable socket by clamps in accordance with Drawing 227-JAN. The measurements on the tube in an oscillating state shall be made with the output line coupled into measuring circuits as per Drawing 227-JAN.

3. No warmup time is required in order to obtain the specified life from the tube. If extended life is desired, a heater warmup period of approximately 1 minute is recommended before the application of the resonator voltage.

4. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0, inspection level II.

5. The cathode lead to be opened shall be cathode (1) in accordance with Method 4229. Reflector current shall be measured after 2 minutes operation, including preheat time.

6. The thermal tuning speed shall be measured in the following manner: The tuner diode voltage and reflector voltage shall be mutually adjusted to produce maximum power under stable conditions of the initial frequency indicated for thermal tuning time (1) or thermal tuning time (2). The tuner diode voltage and reflector voltage shall then be switched simultaneously to the end condition values indicated in the table. The time interval between the instant of switching and the time at which the tube delivers maximum output at the end frequency of the test shall not exceed the specified value.

Test	Condition	Ec and Er conditions	Frequency
Tuning time (1)	Initial	Mutually adjusted for max Po in Mode A at initial frequency.	8,500 MHz ± 0.3%
	End	Ecc = -35 Vdc; Er (Mode A)/max Po at F = 9,660 MHz.	9,600 MHz ± 0.3%
Tuning time (2)	Initial	Mutually adjusted for max Po in Mode A at initial frequency.	9,600 MHz ± 0.3%
	End	Ecc = +15 Vdc; Er (Mode A)/max Po at F = 8,500 MHz.	8,500 MHz ± 0.3%

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NOTES: -Continued

7. Changes from previous issue. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - MU
Air Force - 17
DSA - ES

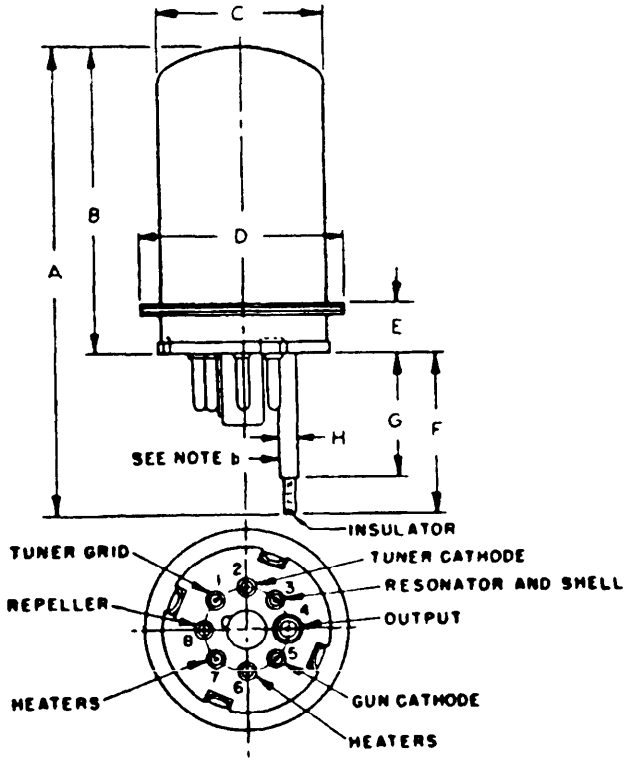
User activities:
Army - WC
Navy - AS, OS, MC, CG, SH
Air Force - 11

Preparing activity: Navy - EC

Agent: DSA - ES

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses (see note c)	
	Minimum	Maximum
Qualification		
B		2.563 (65.10)
Quality conformance inspection, part 1		
F	1.203 (30.56)	1.234 (31.34)
G	.910 (23.11)	.930 (23.62)
H	.135 (3.43)	.145 (3.68)
Quality conformance inspection, part 2		
A		3.797 (96.44)
C		1.281 (32.54)
D	1.500 (38.10)	1.625 (41.28)
E	.375 (9.53)	.438 (11.11)

NOTES:

a. This base shall be capable of being inserted in a gage 1-7/32 (30.96) thick having 7 holes 1/4 (6.35) deep from the top of gage whose diameters are .103 (2.62) for the contact pins. Remaining portion of hole to be clearance approximately 1/64 (.40) larger in diameter and an eighth hole whose diameter .160 (4.06) - 1-7/32 (30.96) deep for the output lead. All holes located on the true centers. Also a center hole having the contour of the pilot but with a clearance of .002 (.05) over the maximum diameter.

b. Nickel (30 M.S.I. silver plate in accordance with QQ-S-365 permissible).

c. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 1. Outline drawing of electron tube type 2K45.

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